

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (Biotechnology) (Credit System) Degree Examinations

May - June 2023

20BTC206 - CLINICAL RESEARCH AND DATA MANAGEMENT

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Outline the history of the Drug Discovery process.	10	L*1	1	2,11
	b) Expand upon the detailed steps involved in the receptor – based approach.	10	L2	1	2,11
2.	a) Discuss in detail about the need for Pharmacological background for drug evaluation.	10	L2	1	2,11
	b) Elaborate on the different methods available for assessing the toxicity.	10	L2	2	2,9
3.	a) Describe about: i) Pharmacokinetics and Pharmacodynamics ii) Bioavailability and Bioequivalence	10	L2	2	2,9
	b) Extrapolate the current methods followed in pharmacovigilance medication errors.	10	L2	2	2,9
Unit – II					
4.	a) Illustrate on Design decisions and the need for Randomization and Blinding.	10	L2	3	2,3,9
	b) Determine the significance of prevalence surveys or the cross-sectional studies in clinical trial.	10	L3	3	2,3,9
5.	a) Elaborate on: Phases of clinical trials-II and Phases of clinical trials-III.	10	L2	3	2,3,9
	b) Prioritize the role of Clinical Research Organization (CRO) and state the principles of clinical data management.	10	L1	4	2,8
6.	a) Derive the data preparation process and data analyses.	10	L4	4	2,8
	b) Demonstrate the critical need of SAS for clinical data management.	10	L3	4	2,8
Unit – III					
7.	a) Enumerate the Good Clinical Practices followed in clinical trials.	10	L1	5	2,8
	b) State the essentiality for strictly follow the clinical trial protocols and follow protocol amendments.	10	L1	5	2,8
8.	a) Outline the Regulatory environment in India- CDCSO, DCGE, ICM.	10	L2	5	2,8
	b) Extend on the Regulatory environment in US-FDA.	10	L2	5	2,8

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
